$h(x) = x^{2/3}(x-4)$

a. Domain = $\mathbb{R}$

b. Intercepts (0,0), (4,0)

c. Asymptotes

1. Horizontal: None since $\lim\limits_{x\to\infty} x^{2/3}(x-4) = \infty$, $\lim\limits_{x\to-\infty} x^{2/3}(x-4) = -\infty$
2. Vertical: None

d. $h'(x) = \dfrac{5x-8}{3x^{1/3}} \Rightarrow$ C.N. $x = 0$, $x = 8/5$

e.

| | | | | | |
|---|---|---|---|---|---|
| h ′ | + | | - | | + |
| h | ↗ | 0 | ↘ | 8/5 | ↗ |

Increasing on $(-\infty, 0) \cup (8/5, \infty)$

Decreasing on $(0, 8/5)$

f. *P* (0,0) is a relative maximum point on the graph
*P* (8/5,-3.283) is a relative minimum point of the graph

g. $h''(x) = \dfrac{2(5x+4)}{9x^{4/3}} \Rightarrow 2^{nd}$ C.N. $x = -4/5$, $x = 0$

h.

| | | | | | |
|---|---|---|---|---|---|
| $h''$ | - | | + | | + |
| $h$ | $\cap$ | -4/5 | $\cup$ | 0 | $\cup$ |

Concave up on $(-\tfrac{4}{5},0)\cup(0,\infty)$

Concave down on $(-\infty,\tfrac{4}{5})$

i. Inflection points: (-4/5,-4.1365)

j. Values

| *x* | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *y* | -26.3 | -20.2 | -14.6 | -9.52 | -5 | 0 | -3 | -3.17 | -2.08 | 0 | 2.924 | 6.604 | 10.98 |

k.